

Robert Hickman

LEAD QUANTITATIVE DEVELOPER

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Employment History

Pythia Sports

London, UK

LEAD QUANTITATIVE DEVELOPER

May 2024 - Present

- Overhaul of data processing and quantitative modelling of racing data
- Development of LLM system to generate horse racing comments
- Optimisation of models through rewrite in compiled Python and Rust
- Management and development of junior quantitative developers

Sporting Group

London, UK

SENIOR QUANTITATIVE ANALYST AND HEAD OF MODEL DATA TEAM

Aug 2023 - May 2024

- Led team of 3 analysts developing data infrastructure and pricing models for quantitative trading desk
- Established robust engineering practices including Git workflows, Poetry package management, and automated linting through GitHub Actions
- Architected and maintained data pipelines across Snowflake and BigQuery databases, with deployments managed through AWS and GCP
- Developed sophisticated customer profiling algorithm combining analytical mathematics with machine learning
- Built Shiny app and REST API infrastructure in C# to serve pricing models to trading desk
- Migrated containerized data collection apps from Jenkins to GitHub Actions, improving deployment efficiency
- Created standardized internal Python libraries for database connectivity, logging, and API management

Kindred Quant Team

London, UK

QUANTITATIVE ANALYST

Aug 2021 - Aug 2023

- Oversaw research and development of pricing models for basketball and ice hockey markets from first principles
- Conducted comprehensive literature reviews to develop theoretical frameworks, implementing production solutions in high-performance Python using NumPy/SciPy
- Optimized Monte Carlo simulations in FORTRAN for performance-critical calculations
- Designed and implemented market risk management system using copula theory to handle correlated sporting events
- Led adoption of software engineering best practices, including CI/CD pipelines and code review processes
- Mentored junior analysts in Python development, Git workflow management, and SQL query optimization
- Authored internal documentation for data modeling approaches and maintained shared statistical libraries

Focaldata

London, UK

DATA SCIENTIST

Jan 2021 - Aug 2021

- Developed statistical modeling pipelines for political polling and market research at early-stage startup
- Built automated sampling and weighting systems using multilevel regression with post-stratification (MRP) in Python
- Orchestrated data pipelines through Airflow and GNU Make
- Implemented reproducible ETL processes using dbt for transforming raw survey data into analysis-ready datasets
- Created comprehensive analysis pipelines combining SQL and dbt transformations to deliver actionable insights
- Published polling analysis in The Times newspaper predicting electoral outcomes

Fleetwood Town Football Club

Fleetwood, UK

ANALYTICS CONSULTANT

Jan 2020 - Dec 2020

- Established club's first data science function, building automated pipelines in Python to process high-frequency event data
- Developed statistical models for player recruitment, including expected goals (xG) modeling and performance forecasting
- Created interactive visualization suite using Plotly and Dash for coaching staff to analyze tactical patterns
- Led analysis for League One season interruption response, developing Bayesian models to predict final league standings
- Built recruitment database and dashboard system combining player performance metrics with scouting reports

Amey Consulting, Strategic Consulting Team

Manchester, UK

DATA SCIENCE INTERN

Jun 2017 - Oct 2017

- Developed spatial regression models in R to predict infrastructure failures across North West England sewage network
- Implemented geospatial analysis using sf and sp packages, incorporating environmental and usage data
- Engineered high-performance algorithm for detecting cyclic deformation in rail track infrastructure
- Optimized implementation from R prototype to production C++ deployment
- Presented technical findings and ROI analysis to stakeholders from National Rail and utility companies

University of Oxford, Dept. of Experimental Psychology

Oxford, UK

RESEARCH ASSISTANT

Jan 2017 - May 2017

- Developed and optimized Python pipelines for generating 3D perceptual stimuli using Blender's Python API
- Created automated testing framework for Amazon Mechanical Turk experiments
- Work contributed to published findings in PNAS (Flesch et al. 2018) on computational models of human perception

Education

University of Cambridge

Cambridge, UK

PHD NEUROSCIENCE AND ECONOMICS

2017 - 2024

- Led computational research combining machine learning and economic theory to model neuronal basis of economic decision making
- Developed MATLAB suite for real-time behavioral experiments, integrating with C++ controlled National Instruments hardware
- Built support vector machines to decode neural activity patterns and implemented reinforcement learning models
- Work published in Nature Communications (Hill, Hickman et al. 2024)

Institute of Molecular Pathology, Vienna Biocenter

Vienna, Austria

PHD MOLECULAR BIOLOGY (INCOMPLETE)

2013 - 2016

- Research in biophysics and protein engineering
- Mentored Masters student in quantitative analysis
- Left program after 3 years for personal reasons

St Hugh's College, University of Oxford

Oxford, UK

BA PHYSIOLOGICAL SCIENCES (2.1)

2010 - 2013

- Focus on neuroscience and neural basis of decision making
- Analyzed neural networks in MATLAB to predict future auditory scenes, compared to physiological data from ferrets

Technical Skills

Programming Languages: Python (NumPy, SciPy, pandas, scikit-learn, statsmodels, numba, Poetry), R (statistical analysis, CRAN packages, Rcpp), C#, Rust, C++, FORTRAN, MATLAB, Stan

Data Infrastructure: PostgreSQL, Snowflake, BigQuery, MongoDB, dbt, Apache Kafka, SQL optimization

Development Practices: Git/GitHub, CI/CD (GitHub Actions), Docker, GNU Make, package development, code review

Cloud Platforms: AWS, GCP, Apache Airflow

Visualization: Shiny, Dash, Plotly, Jupyter, matplotlib, ggplot2

Statistical Methods: Copula theory, multilevel regression, Monte Carlo simulation, Bayesian modeling

Publications & Awards

Hill, Hickman, Al-Mohammad, Stasiak, and Schultz (2024). Dopamine neurons encode trial-by-trial subjective reward value in an auction-like task. *Nature Communications*

Meers, Hickman, and Leeper (2019). ggparliament: a ggplot2 extension for parliament plots in R. *Journal of Open Source Software*

Cambridge AI & Biological Cognition Symposium (2019). Poster Prize - 1st place

Boehringer Ingelhem Fonds PhD Fellowship (2014). 3-year fellowship for outstanding junior scientists

Vienna BioCenter PhD Fellowship (2013). Fully funded admission

Wellcome Trust Masters Scholarship (2013). £12,000 for most promising Oxford MSc Pharmacology applicant